

PHOTONICS Research

Self-aligned dual-beam superresolution laser direct writing with a polarization-engineered depletion beam

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A fiber-based, self-aligned dual-beam laser direct writing system with a polarization-engineered depletion beam is designed, constructed, and tested. This system employs a vortex fiber to generate a donut-shaped, cylindrically polarized depletion beam while simultaneously allowing the fundamental mode excitation beam to pass through. This results in a co-axially self-aligned dual-beam source, enhancing stability and mitigating assembly complexities. The size of the central dark spot of the focused cylindrical vector depletion beam can be easily adjusted using a simple polarization rotation device. With a depletion wavelength of 532 nm and an excitation wavelength of 800 nm, the dual-beam laser direct writing system has demonstrated a single linewidth of 63 nm and a minimum line spacing of 173 nm. Further optimization of this system may pave the way for practical superresolution lithography that surpasses the diffraction limit. © 2024 Chinese Laser Press

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1. INTRODUCTION

Developments in the semiconductor industry have intensified the demand for high-resolution, small-featured structures at micro and nanoscales [1–3]. Laser direct writing (LDW) addresses many nanostructure requirements due to its simplicity and three-dimensional fabrication capability. However, LDW is limited by the diffraction limit in its ability to create high-resolution nanostructures [4]. The stimulated emission depletion (STED) technique, proposed in 1994 [5], inspired the development of stimulated emission depletion laser direct writing (STED-LDW) as a novel strategy for overcoming LDW limitations [6]. This advanced LDW system has broad applications in micro and nanofabrication, including supersurfaces [7,8], metamaterials [9,10], photonic crystals [11], and photonics chips [12].

The size of the central dark spot of the depletion beam significantly influences the feature size and resolution of the STED-LDW system. In general, a smaller dark spot size increases the achievable resolution, enabling the production of finer patterns. Traditional STED-LDW systems attempt to reduce dark spot size by intensifying the depletion beam [13]. For instance, Wollhofen *et al.* achieved a feature size of 55 nm in 2013 by amplifying the depletion beam's power [14]. Similarly, Kuang *et al.* in 2023 established 35 nm nanowires using a

spatial light modulator in the single color peripheral photoinhibition lithography system [15]. Other methods involve modulating the phase or shape of the depletion beam. In 2019, He *et al.* designed a rod-shaped incident depletion beam, achieving a linewidth of 45 nm [16]. Additionally, the depletion wavelength plays a role. Gu *et al.* in 2013 used a 375 nm depletion wavelength to fabricate 9 nm features [17]. However, these approaches often involve complex alignment steps and neglect the impact of depletion beam polarization on the dark spot size [18,19].

This work presents what we believe is a novel design for a convenient coaxial dual-beam source that enhances system stability by coupling excitation and depletion beams into a vortex fiber [20,21]. This approach simplifies alignment and reduces complexity. Additionally, a simple polarization rotation device allows for adjusting the size of the central dark spot of the focused cylindrical vector depletion beam. This adjustment mitigates potential damage to the photoresist caused by increased depletion beam power. Using a depletion wavelength of 532 nm, an excitation wavelength of 800 nm, and photoresist, we achieved a single linewidth of 63 nm with a minimum line spacing of approximately 173 nm. This demonstrates the feasibility of the fiber-based self-aligned dual-beam LDW system and the adjustable dark spot size. Further optimization of

this system holds promise for even smaller feature sizes in STED-LDW applications.

2. EXPERIMENTAL SETUP

The experimental setup is shown in Fig. 1(a). An excitation beam (Mai Tai, Spectral Physics) with central wavelength at 800 nm, pulse width of 45 fs, and repetition rate of 80 MHz is used in the system. The power of the excitation beam can be adjusted through a half-wave plate (HWP1) and a polarizing beam splitter (PBS). Another half-wave plate (HWP2) is employed to rectify the polarization induced by the double HWPs after the fiber tail for the excitation beam. A continuous-wave solid-state laser with a wavelength of 532 nm (MGL-FN-532, Changchun New Industries Optoelectronics Technology) is used as the depletion. The polarization state of the depletion beam prior to coupling into the fiber is adjusted by a half-wave plate (HWP3) and a quarter-wave plate (QWP), facilitating the polarization shaping after depletion beam is emitted from the fiber tail. HWP3 and QWP are used only to change the polarization state of the depletion beam to either circular polarization with solid spot or azimuthal polarization in annular shape. A depletion beam with a circularly polarized solid spot is used for the complete erasure experiment, while the depletion beam with azimuthal polarization in annular shape is used for STED-type writing. The excitation and depletion beams are combined through a dichroic mirror (DM), and then simultaneously coupled into the vortex fiber by a lens. The fiber retains the excitation beam as a Gaussian beam with linear polarization and slightly broadening the pulse width, while transforming the depletion beam into either a donut shape azimuthally polarized beam or a solid circularly polarized beam. Such a vortex fiber provides a coaxial dual-beam source for the LDW system. A linear polarizer (LP) can be used to adjust the polarization states of dual beams, enabling precise control over the intensity distribution of depletion beam. A combination of broadband half-wave plates (HWP4 and HWP5, LbTek) is utilized for depletion beam polarization shaping, offering straightforward

polarization control over the depletion beam to affect the dark spot size of the depletion beam and the LDW results. Finally, a high numerical aperture objective lens (OL2; UPlanSApo 60 $\times$ /1.35, Olympus) focuses the dual beams within the photoresist, actualizing three-dimensional fabrication. The piezoelectric ceramic translation stage (P-733.3 XYZ, Physik Instrumente) governs the speed and direction of LDW. As illustrated in Fig. 1(b), the double HWPs can convert the azimuthally polarized beam into a generalized cylindrically symmetric vector beam, the rotation of which is determined by the angle $\Delta\varphi$ between the fast axes of the two HWPs. By merely rotating one of the HWPs, the fast axis $\Delta\varphi$ and the polarization angle φ can be altered, where $\Delta\varphi = \varphi/2$ [22,23]. The photoresist is prepared by blending the monomer IP-L780 (Nanoscribe) and the photoinitiator DETC with a doping ratio of 0.5% (mass fraction), which is then left to rest for 5 h post-mixing. The monomer IP-L780 by Nanoscribe is a negative photoresist and 200 μ L of mixed photoresist on the glass substrate is used for processing. After LDW, the sample undergoes development in isopropanol for 5 min followed by a rinse with acetone for 1 min. To inspect the fabricated samples in a scanning electron microscope (SEM), 30 s of gold sputtering is applied to make the sample surface conductive. The sample surface morphology, linewidth, and line spacing are gauged from the SEM images. Atomic force microscope (AFM) images of the sample are also used to analyze resolution.

3. EXPERIMENTAL RESULTS

The effects of the average power of the excitation beam at the entrance pupil, the writing speed, and the polarization on the linewidth were investigated at first in order to analyze the optimal system parameters for the fiber-based STED-LDW system exposure. HWP2 is fixed to maintain consistent excitation beam polarization with the stage movement, and the stage speed is set at 400 μ m/s. By rotating HWP1, the excitation beam power can be adjusted, with laser writing initiated at 7.8 mW. The average power of the excitation beam at

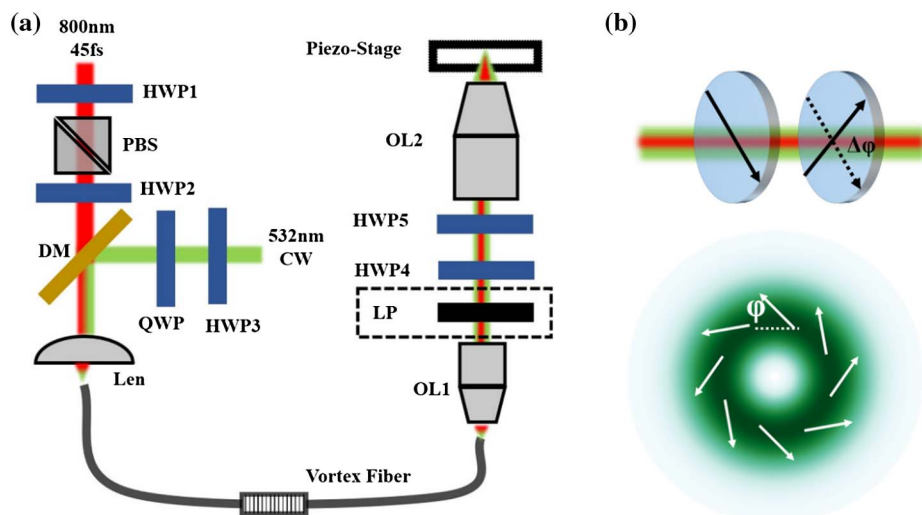


Fig. 1. (a) Schematic diagram of the optical path of the STED-LDW system (HWP, half-wave plate; PBS, polarizing beam splitter; DM, dichroic mirror; LP, polarizer); (b) diagram of the double HWPs and the cylindrically symmetric vector beam angle.

the pupil entrance is then subsequently increased, as illustrated in Fig. 2(a). At a constant speed, with the decrease in average power at the pupil entrance, the processed feature size continues to diminish until the writing features are no longer viable. The smallest and most distinct feature size occurred at an average power of 7.8 mW, while the linewidth reaches 111 nm. As demonstrated in Fig. 2(b), with the excitation beam polarization parallel to the writing direction and the average power of the excitation beam at the pupil entrance being 7.8 mW, the linewidth gradually decreases as the writing speed increases. An excessively high speed might result in broken lines. Finally, the influence of polarization on linewidth is examined. HWP1 is fixed to set the excitation beam power at 7.8 mW with a speed of 400 $\mu\text{m/s}$. The single-pulse energy at the focal point is about 0.975 nJ with laser fluence of $\sim 1.4 \text{ J/cm}^2$. HWP2 is rotated so that the angle between the polarization of the excitation beam and the translation direction is adjusted to be 0° and 45° , respectively. As shown in Fig. 2(c), as the angle between the polarization direction and the translation direction increases, the linewidth gradually increases. Ultimately, when the polarization of the excitation beam is set to be parallel to the writing direction, the minimum linewidth increases to 122 nm. Based on these results, we set the excitation beam average power to be 7.8 mW with polarization parallel to the writing direction, and the writing speed is set to 400 $\mu\text{m/s}$.

Subsequently, the depletion effect with the IP-L780 photoresist is evaluated. HWP3 and QWP are adjusted to produce a solid circularly polarized depletion beam from the fiber end. The resulting patterns in Fig. 3 are obtained with the excitation beam always activated, while the depletion beam is only switched on for a specific time interval. The individual red lines represent processing with only the excitation beam, and the parts with green lines represent the simultaneous activation of excitation beam and depletion beam. Different doping ratios of the photoinitiator are employed for the processing. For undoped photoresist, the position processed by dual-beam exposure does not exhibit an erasure effect. However, when the photoresist contained 0.5% (mass fraction) of the photoinitiator, a pronounced erasure effect appeared after development, as depicted in Fig. 3(a). Under the same conditions, the depletion beam is a radially polarized beam, as shown in Fig. 3(b). This demonstrates that the photoresist exhibits a complete erasure phenomenon under the fiber-based dual-beam exposure and the change in polarization state of the depletion beam can still sustain the erasure phenomenon.

The fiber-based STED-LDW system can also realize the function of spatial STED-LDW. HWP3 and QWP are adjusted to emit a donut-shaped cylindrically polarized depletion beam from the fiber. Both the excitation beam and the depletion beam are simultaneously activated for processing

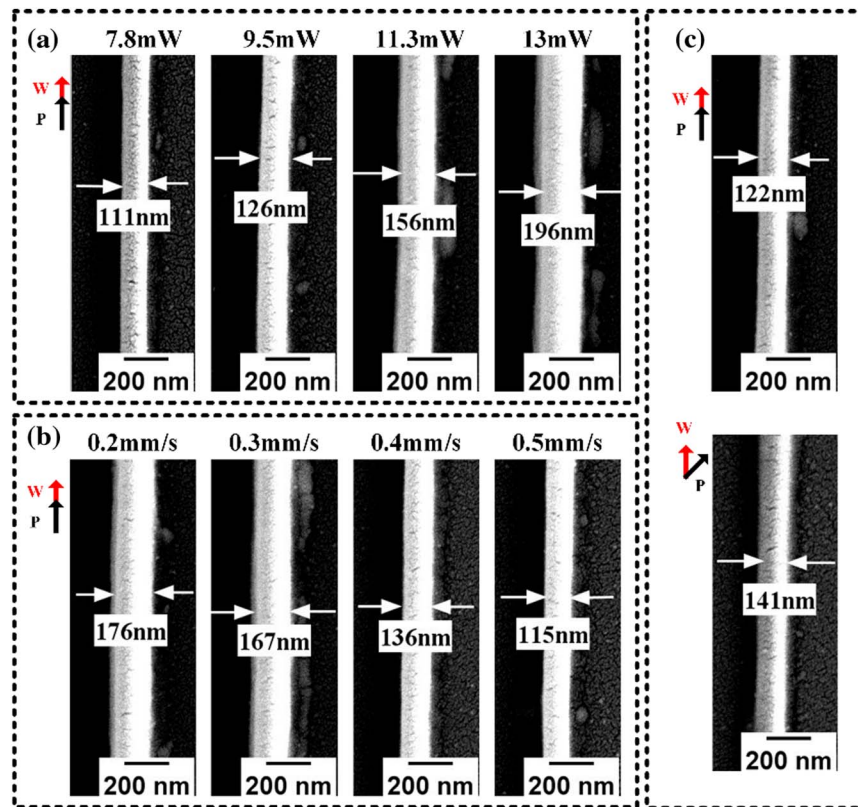


Fig. 2. Effects of average power, speed, and polarization on linewidth. (“W” represents the writing direction and “P” represents the polarization direction during the excitation beam processing.) (a) shows the effect of excitation beam on linewidth, with the linewidth increasing successively with the increase in power; (b) shows the effect of LDW speed on linewidth, where an increase in speed leads to a decrease in linewidth; (c) shows the effect of excitation beam polarization on the processing results.

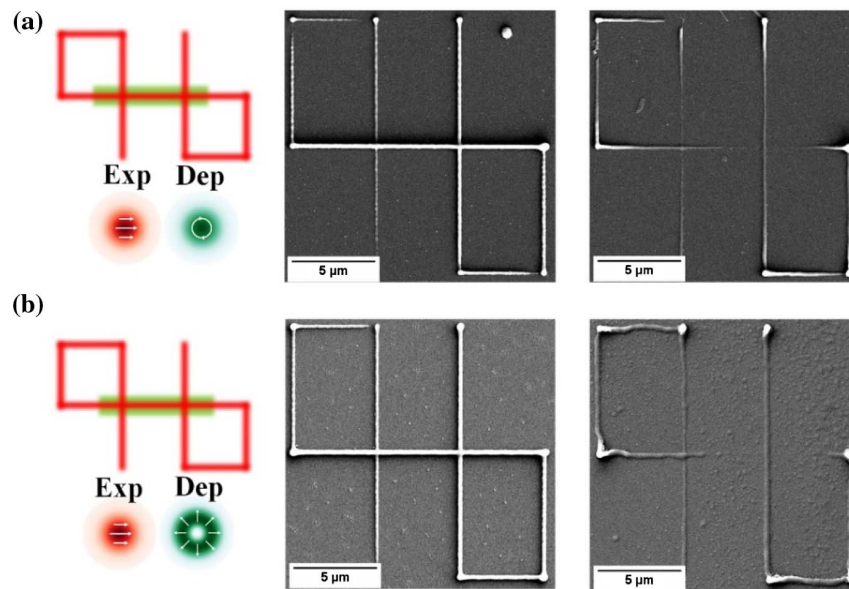


Fig. 3. Erasure effect with different depletion beams. (a) Test patterns with solid circularly polarized depletion beam without DETC doping and with 0.5% (mass fraction) DETC doping; (b) test patterns for radially polarized depletion beam under the same conditions as (a).

with the depletion beam power incrementally increased by 3 mW in each case. The SEM images are presented in Fig. 4(a). As the depletion beam power increases, the depletion effect on the edge of the excitation beam processing also increasingly intensifies. At a depletion beam power of 3 mW, the dual-beam processing begins to exhibit depletion effect. When the depletion power reaches 12 mW, a minimum linewidth of 72 nm can be realized. However, if the depletion beam power is too high, phenomena such as complete erasure or

widening of the single line might occur, rendering unreliable smaller feature sizes. This is because, even though increasing the depletion power may further reduce the photopolymerization area in principle, the depletion energy at the edge during processing also increases dramatically, making further reducing the linewidth impractical. When the LP is placed in front of the double HWP's, the LP is rotated to make the intensity distribution of the depletion beam present a bilobed shape, with the direction as shown in Figs. 4(b)–4(d). The excitation beam

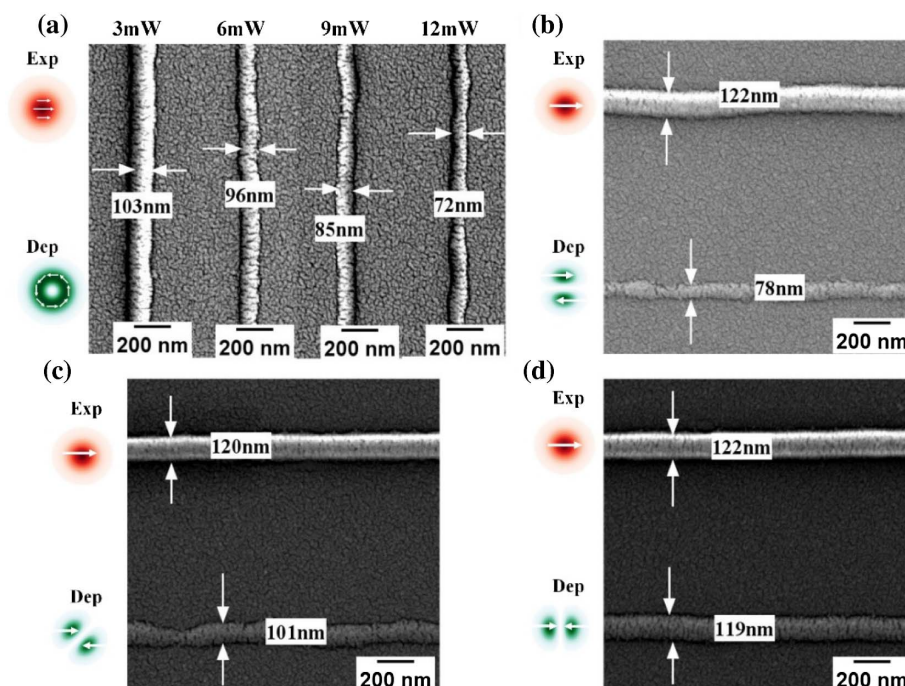


Fig. 4. (a) Feature size of the nanowire processed by excitation beam under different depletion beam power. (b)–(d) show the processing results with different depletion beam intensity distributions, while other conditions remain the same.

power is kept around 8 mW to ensure that the linewidth of single-beam processing is maintained at around 120 nm. Subsequently, with a power of 9 mW, a depletion beam with a different light intensity distribution is added, with the feature size shown in Figs. 4(b)–4(d). When the intensity distribution of depletion beam is aligned with the processing direction, the lateral inhibition is most pronounced. When the direction is perpendicular, longitudinal inhibition is the dominant phenomenon.

Finally, without altering the power, phase, and wavelength of the excitation beam, the polarization of the depletion beam is adjusted merely by rotating the HWP5, and the LDW performance is investigated. The depletion beam power is set at 9 mW at which the depletion phenomenon is clearly observed without damaging the photoresist. HWP5 at the fiber tail is rotated to adjust the angle between the fast axes of the two HWPs, accomplishing the adjustment of the incident polarization state of the depletion beam at the pupil entrance. Its polarization and intensity distribution are illustrated in Fig. 5(a), where the angle for the cylindrical vector polarization from the azimuthal direction is set to be 0° , 12° , 24° , and 36° , respectively. By adjusting the polarization state of the excitation beam in advance with HWP2 before coupling into the fiber, a consistent polarization state of the excitation beam is maintained. The focal fields of the depletion beam and the excitation beam are simulated using the widely adopted Richard–Wolf diffraction formula through MATLAB, and the numerical aperture is set to 0.95 [24], as shown in Fig. 5(b). The red color represents the simulated focus of the excitation beam, while the green color represents the focus of the depletion beam under different polarization states. With the change of cylindrical vector polarization angle, the central dark spot size of the depletion

beam focus gradually decreases, and the shape of focus transforms from hollow to solid. From the projections in three cross sections, the overlap between the two beams gradually increases as the polarization of the depletion beam changes. This means that the excitation beam available for processing is gradually decreasing, leading to the experimental results shown in Figs. 5(c) and 5(d). While maintaining the power of the depletion beam and the excitation beam, single-beam processing of nanowires has been achieved with the feature size maintained at 110 nm; the feature size of dual-beam processing under different depletion beam polarization states is summarized and shown in Fig. 5(c). The linewidth decreases while the dark center of the depletion beam fills up; the fabricated feature size also gradually reduces and a minimum linewidth of 63 nm is ultimately achieved. Compared with the method of increasing the depletion beam power to reduce the feature sizes, the method of polarization adjustment produces better resolution and tolerance of the photoresist with very easy operation. Figure 5(d) presents the smallest resolvable distance of the fabricated lines obtained by AFM under the same conditions, which is approximately 173 nm, significantly improved compared to the prepolarization control. These results clearly demonstrate advantages of this polarization-controlled fiber-based dual-beam LDW system for optimizing resolution and reducing feature size.

4. DISCUSSIONS AND CONCLUSIONS

In this work, we utilize a vortex fiber to conveniently achieve co-axial alignment between the excitation and depletion beams in an LDW system, thereby improving the equipment stability. Comparing with other dual-beam systems, this approach

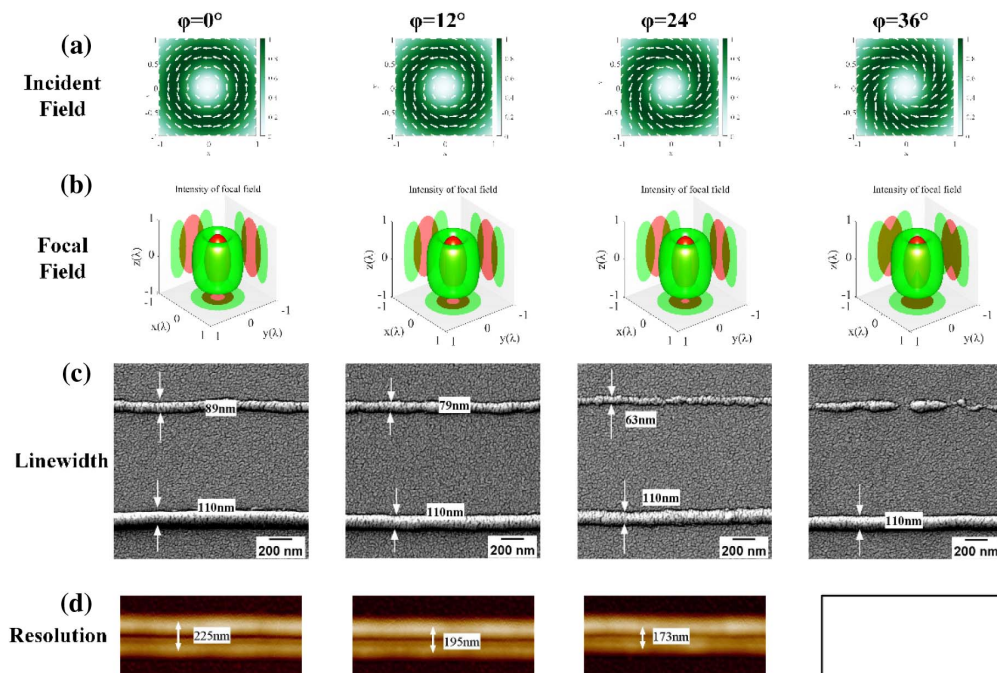


Fig. 5. (a) The beam intensity and polarization distribution of the depletion beam at the pupil entrance; (b) the focal field distributions of the dual beams after focusing with a high numerical aperture objective lens; (c) the feature size under different depletion beam polarizations; (d) the resolution under different depletion beam polarizations.

simplifies the tedious alignment procedures in STED-LDW and allows easy switching between donut-shaped and Gaussian states via merely altering the polarization state of the depletion beam. This design also makes it possible to accomplish dual-beam depletion processing without extensive analysis of the focal point location, thus making the implementation of STED-LDW more straightforward. The size of the dark spot at the center of the depletion beam significantly impacts the performance in STED-LDW. Given that the donut-shaped mode emitted from the fiber tail is cylindrically polarized, it can be switched to other vector beams using a double wave plate setup, thereby facilitating simple adjustments of the hollow center size of the focused depletion beam. However, due to the notable wavelength differentiation between the excitation and the depletion, chromatic aberration in the axial direction from the focusing lens cannot be ignored, which affects the resolution and impedes further improvement of the system performance. Challenges arising from chromatic aberration are common to the dual-beam LDW system, which are not specific to the fiber-based STED-LDW system. However, through utilizing a super-apochromatic objective during the focusing processes, the positional discrepancies that occur after dual-beam focusing are reduced in the fiber-based STED-LDW system. Furthermore, based on the principle of STED, the overlapping part of the dual beams during processing is more important. The method used in the fiber-based STED-LDW system involves hiding the focus of the excitation beam, which does not overlap with the focus of the depletion beam, in the glass substrate during the processing to minimize the effect arising from axial chromatic aberration. In the future, employing a light source with a lesser wavelength difference between the two beams, such as a continuous beam and pulsed beam of nearly the same wavelength, will further improve the processing quality. Additionally, using IP-L780 combined with DETC might diminish the depletion effect [25], necessitating a higher processing speed and impeding the attainment of smaller feature sizes. Hence, it might be worth considering the use of a photoresist with a superior STED effect in such an LDW system in the future.

In summary, we present a novel fiber-based, self-aligned dual-beam LDW system with a polarization-engineered depletion beam. This system utilizes a vortex fiber to simultaneously generate a donut-shaped, cylindrically polarized depletion beam and transmit the fundamental mode excitation beam, achieving inherent self-alignment that enhances stability and reduces assembly complexity. A double-HWP polarization rotator easily adjusts the central dark spot of the focused depletion beam. Currently, this system achieves a single-line linewidth of 63 nm and a minimum line spacing of 173 nm, demonstrating subdiffraction-limit resolution and feature sizes. Further optimization promises even smaller feature sizes and paves the way for practical superresolution photolithography with simplified alignment, enhanced stability, improved performance, increased throughput, and reduced photoresist requirements.

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Data Availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

1. J. Fischer, J. B. Mueller, A. S. Quick, *et al.*, "Exploring the mechanisms in STED-enhanced direct laser writing," *Adv. Opt. Mater.* **3**, 221–232 (2015).
2. M. He, Z. Zhang, C. Cao, *et al.*, "3D sub-diffraction printing by multi-color photoinhibition lithography: from optics to chemistry," *Laser Photonics Rev.* **16**, 2100229 (2022).
3. J. Chen, G. Chen, and Q. Zhan, "Self-aligned fiber-based dual-beam source for STED nanolithography," *Chin. Opt. Lett.* **19**, 072201 (2021).
4. A. Marino, J. Barsotti, G. De Vito, *et al.*, "Two-photon lithography of 3D nanocomposite piezoelectric scaffolds for cell stimulation," *ACS Appl. Mater. Interfaces* **7**, 25574–25579 (2015).
5. S. W. Hell and J. Wichmann, "Breaking the diffraction resolution limit by stimulated emission: stimulated-emission-depletion fluorescence microscopy," *Opt. Lett.* **19**, 780–782 (1994).
6. M. Deubel, G. Von Freymann, M. Wegener, *et al.*, "Direct laser writing of three-dimensional photonic-crystal templates for telecommunications," *Nat. Mater.* **3**, 444–447 (2004).
7. A. Accardo, M.-C. Blatché, R. Courson, *et al.*, "Two-photon lithography and microscopy of 3D hydrogel scaffolds for neuronal cell growth," *Biomed. Phys. Eng. Express* **4**, 027009 (2018).
8. T. Gissibl, S. Thiele, A. Herkommer, *et al.*, "Two-photon direct laser writing of ultracompact multi-lens objectives," *Nat. Photonics* **10**, 554–560 (2016).
9. C. Yuan, J. Liu, T. Jia, *et al.*, "Super resolution direct laser writing in ITX resist inspired by STED microscopy," *J. Nonlinear Opt. Phys. Mat.* **23**, 1450015 (2014).
10. D. Zhu, L. Xu, C. Ding, *et al.*, "Direct laser writing breaking diffraction barrier based on two-focus parallel peripheral-photoinhibition lithography," *Adv. Photonics* **4**, 066002 (2022).
11. E. D. Lemma, B. Spagnolo, M. De Vittorio, *et al.*, "Studying cell mechanobiology in 3D: the two-photon lithography approach," *Trends Biotechnol.* **37**, 358–372 (2019).
12. A. Frölich, J. Fischer, T. Zebrowski, *et al.*, "Titania woodpiles with complete three-dimensional photonic bandgaps in the visible," *Adv. Mater.* **25**, 3588–3592 (2013).
13. J. Lölsberg, A. Cinar, D. Felder, *et al.*, "Two-photon vertical-flow lithography for microtube synthesis," *Small* **15**, 1901356 (2019).
14. R. Wollhofen, J. Katzmann, C. Hrelescu, *et al.*, "120 nm resolution and 55 nm structure size in STED-lithography," *Opt. Express* **21**, 10831–10840 (2013).
15. M. He, Z. Zhang, C. Cao, *et al.*, "Single-color peripheral photoinhibition lithography of nanophotonic structures," *Photonix* **3**, 25 (2022).
16. X. He, T. Li, J. Zhang, *et al.*, "STED direct laser writing of 45 nm width nanowire," *Micromachines* **10**, 726 (2019).
17. Z. Gan, Y. Cao, R. A. Evans, *et al.*, "Three-dimensional deep sub-diffraction optical beam lithography with 9 nm feature size," *Nat. Commun.* **4**, 2061 (2013).
18. D. Xue, Y. Wang, J. Zhang, *et al.*, "Projection-based 3D printing of cell patterning scaffolds with multiscale channels," *ACS Appl. Mater. Interfaces* **10**, 19428–19435 (2018).
19. C. Ding, D. Zhu, C. Ma, *et al.*, "Method for active spatial alignment and stabilization of laser beams in multi-beam systems," *Chin. Opt. Lett.* **21**, 031201 (2023).



20. S. Ramachandran and P. Kristensen, "Optical vortices in fiber," *Nanophotonics* **2**, 455–474 (2013).
21. L. Yan, P. Kristensen, and S. Ramachandran, "Vortex fibers for STED microscopy," *APL Photonics* **4**, 022903 (2019).
22. Q. Zhan and J. R. Leger, "Focus shaping using cylindrical vector beams," *Opt. Express* **10**, 324–331 (2002).
23. Q. Zhan, "Cylindrical vector beams: from mathematical concepts to applications," *Adv. Opt. Photonics* **1**, 1–57 (2009).
24. J. Chen, C. Wan, L. J. Kong, *et al.*, "Tightly focused optical field with controllable photonic spin orientation," *Opt. Express* **25**, 19517–19528 (2017).
25. C. Cao, Y. Qiu, L. Guan, *et al.*, "Dip-in photoresist for photoinhibited two-photon lithography to realize high-precision direct laser writing on wafer," *ACS Appl. Mater. Interfaces* **14**, 31332–31342 (2022).